

ECEN 215, Spring 2024

Electrical Circuit Theory

Class 7: Capacitor, Inductor, Transient

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OUTLINE



- Capacitor
 - Inductor
 - Transient circuit analysis
-

Ch3: Intro



In this chapter, we consider energy storage elements. Similar to resistors these elements are passive; that is, they do not supply power. This chapter describes the relationship between voltage, current, power and energy for inductors and capacitors. In addition, it explains the effect of connecting these components in series and in parallel.

The Capacitor

A capacitor is an electrical component that consists of two conductors separated by an insulator. When a voltage is applied between the two conductors, a positive charge is deposited on one conductor and a negative charge on the other. The amount of charge stored in the conductors is proportional to the voltage.

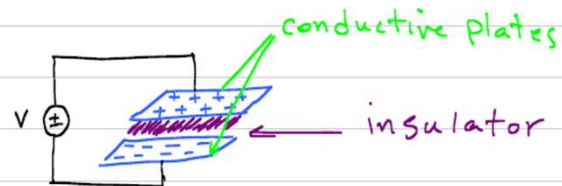
$$q = C V$$

C : capacitance in farads (F)

V : voltage (V)

q : charge in coulombs

$$1 \text{ Farad} = \frac{1 \text{ Coulomb}}{\text{Volt}}$$



Ch3: Capacitor



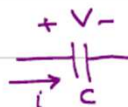
$$C = \frac{\epsilon A}{d}$$

A = Area of each plate

d = distance between the plates

ϵ = permittivity of the dielectric material

$$q = CV$$



$$\frac{dq}{dt} = C \frac{dV}{dt}$$

$$i = C \frac{dV}{dt}$$

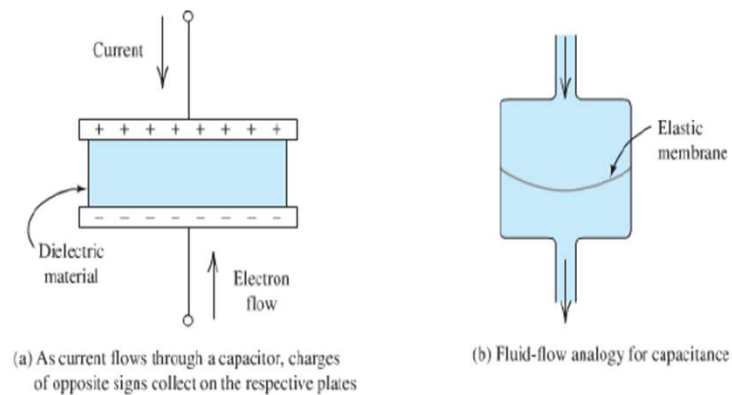
(if i is in the direction of voltage drop)

Ch3: Capacitor



Fluid flow analogy for a capacitor

In terms of fluid flow analogy, a capacitor represents a reservoir with an elastic membrane separating the inlet and outlet. As the fluid flows into the inlet, the membrane is stretched, creating a force (analogous to capacitor voltage) that opposes further flow. The displaced fluid volume is analogous to the charge stored in the capacitor.



Ch3: Capacitor



voltage in terms of current

$$i = c \frac{dv}{dt}$$

$$i dt = c dv$$

$$\int_{t_0}^t i dt = \int_{V(t_0)}^{V(t)} c dv$$

$$V(t) = \frac{1}{c} \int_{t_0}^t i dt + V(t_0)$$

(if i is in the direction of voltage drop)

Ch3: Capacitor



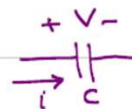
$$C = \frac{\epsilon A}{d}$$

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$$q = CV$$



$$\frac{dq}{dt} = C \frac{dV}{dt}$$

$$i = C \frac{dV}{dt}$$

(if i is in the direction of voltage drop)

Ch3: Capacitor



Power and Energy

$$P = vi = c v \frac{dv}{dt}$$

$$w_{(t)} = \int_{t_0}^t P dt = c \int_{t_0}^t v \frac{dv}{dt} \cdot dt = \frac{1}{2} c v^2 \Big|_{t_0}^t = \frac{1}{2} c v_{(t)}^2 - 0$$

$$w_{(t)} = \frac{1}{2} c v_{(t)}^2$$

Important properties of capacitors

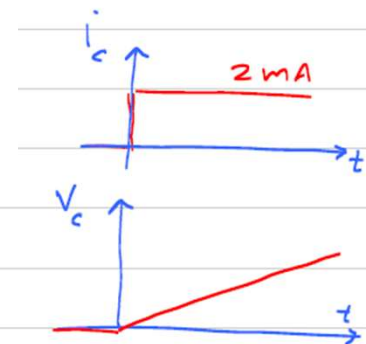
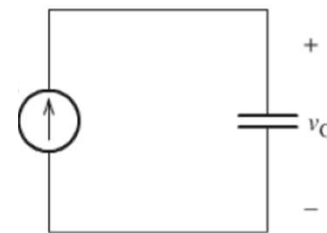
1. A capacitor behaves as an open circuit when its voltage is constant.
2. The voltage across a capacitor cannot change instantaneously. (Capacitor voltage is continuous).

Example1



Ex. At $t=0$, a 2-mA source is applied to an uncharged 2 micro-farad capacitor. Determine

- a) the capacitor voltage at $t=30$ ms.
- b) the energy stored in the capacitor at $t= 30$ ms.



Example1



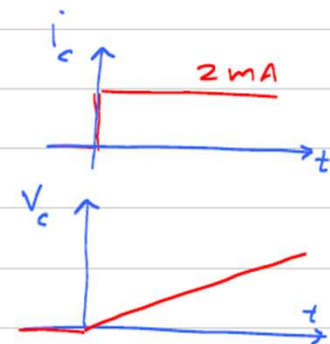
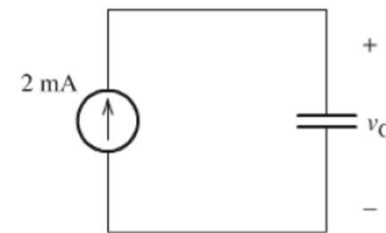
Ex. At $t=0$, a 2-mA source is applied to an uncharged 2 micro-farad capacitor. Determine

a) the capacitor voltage at $t=30$ ms.

b) the energy stored in the capacitor at $t= 30$ ms.

$$\begin{aligned} \text{a) } V &= \frac{1}{C} \int_0^{.03} i \, dt + V_0 \\ V &= \frac{1}{2 \times 10^{-6}} \int_0^{.03} .002 \, dt + 0 \\ &= \frac{1}{2 \times 10^{-6}} \left(.002 t \Big|_0^{.03} \right) = 30 \text{ V} \end{aligned}$$

$$\text{b) } W = \frac{1}{2} C V^2 = \frac{1}{2} (2 \times 10^{-6}) (30)^2 = 0.9 \text{ mJ}$$

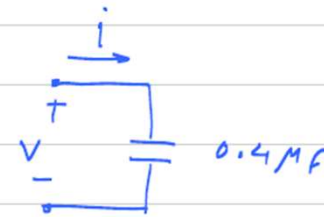


Example2



Ex: The voltage of a 0.4 microfarad capacitor is:

$$V = \begin{cases} 25 \text{ V} , & t < 0 \\ A_1 t e^{-1500t} + A_2 e^{-1500t} , & t \geq 0 \end{cases}$$



and the current at $t=0$ is 90 mA.

- Find the initial energy stored in the capacitor.
- Find A_1 and A_2 .

Example2



$V_c(t=0^-) = V_c(t=0^+)$, capacitor voltage is continuous.

$$a) W = \frac{1}{2} C V_o^2$$

$$= \frac{1}{2} (1.4 \times 10^{-6}) (25)^2 = \underline{125 \mu J}$$

$$b) \left(A_1 t e^{-1500t} + A_2 e^{-1500t} \right) \Big|_{t=0} = 25$$

$$\underline{A_2 = 25}$$

$$i = C \frac{dV}{dt} = (0.4 \times 10^{-6}) \frac{d}{dt} \left[A_1 t e^{-1500t} + 25 e^{-1500t} \right]$$

$$= 0.4 \times 10^{-6} \left[A_1 (e^{-1500t} - 1500t e^{-1500t}) - 25 \times 1500 e^{-1500t} \right] \Big|_{t=0} = 90 \text{ mA}$$

$$\underline{A_1 = 262.5 \frac{\text{KV}}{\text{s}}}$$

Transient Circuit Analysis

- So far, we have been solving for voltage, current, power, without regard to time. This is fine for *steady-state dc* or for *instantaneous* situations.
- Now, we will look at how circuits change over time.
- Two main elements will be considered: capacitors and inductors. Both of these have an equation that depends on time.
 - Capacitors store energy in the form of an electric field between two charged plates (sort of like it is “storing voltage”)
 - Inductors store energy in the form of a magnetic field around a coil of wire (sort of like it is “storing current”)
- We'll solve these (1) steady-state before the switch, (2) instantaneously after the switch, (3) steady-state after the switch

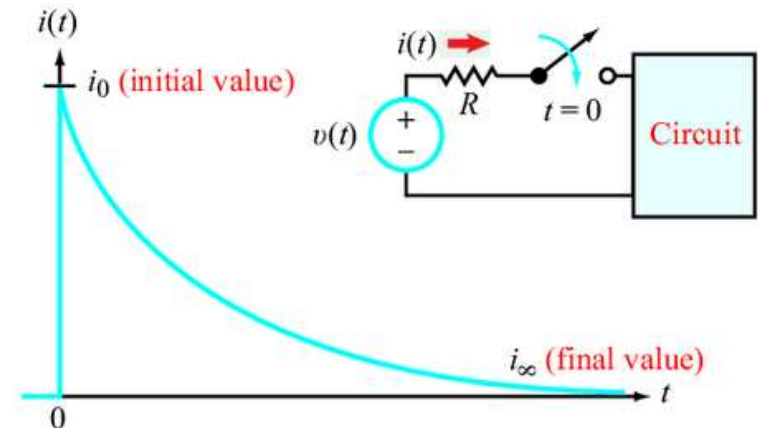
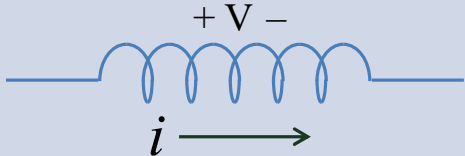
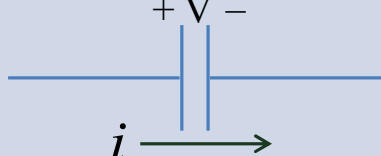


Image from Ulaby, Maharbiz, Furse

Introduction to Inductors and Capacitors

- Inductors and capacitors are passive devices which store (and release) energy (energy is not created/generated by a passive device).

	Inductor	Capacitor
Circuit Symbol		
Fundamental I/V relationship	$v = L \frac{di}{dt}$	$i = C \frac{dv}{dt}$
Units	L=inductance (Henrys)	C=capacitance (Farads)



Properties of Inductors and Capacitors

- From the fundamental I/V relationship we can infer several properties of inductors and capacitors.

	Inductor	Capacitor
Fundamental I/V relationship	$v = L \frac{di}{dt}$	$i = C \frac{dv}{dt}$
Instantaneous changes	The current in an inductor cannot change instantaneously. To do so would cause an infinite voltage to appear.	The voltage across a capacitor cannot change instantaneously. To do so would cause an infinite current to flow.
Steady State (when things are not changing with time)	When a circuit with an inductor reaches steady state, the voltage across the inductor will be zero. → An inductor acts like a short circuit in steady state.	When a circuit with a capacitor reaches steady state, the current through the capacitor will be zero. → An capacitor acts like an open circuit in steady state.

Inverse to Fundamental I–V Relationships

- From the fundamental I–V relationships we can also derive inverse relationships.

Inductors

$$v = L \frac{di}{dt}$$

$$\int_{t_o}^t v(u) du = L \int_{t_o}^t \frac{di}{du} du$$

$$\int_{t_o}^t v(u) du = L(i(t) - i(t_o))$$

$$i(t) = i(t_o) + \frac{1}{L} \int_{t_o}^t v(u) du$$

Capacitors

$$i = C \frac{dv}{dt}$$

$$\int_{t_o}^t i(u) du = C \int_{t_o}^t \frac{dv}{du} du$$

$$\int_{t_o}^t i(u) du = C(v(t) - v(t_o))$$

$$v(t) = v(t_o) + \frac{1}{C} \int_{t_o}^t i(u) du$$

Power and Energy in Inductors and Capacitors



- From the fundamental I–V relationship we can infer several properties of inductors and capacitors.

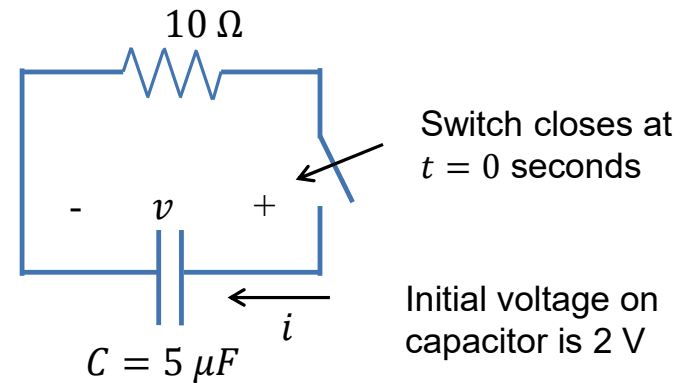
	Inductor	Capacitor
Fundamental I–V relationship	$v = L \frac{di}{dt}$	$i = C \frac{dv}{dt}$
Power	$p(t) = i(t)v(t) = Li(t) \frac{di}{dt}$ $= \frac{L}{2} \frac{d}{dt} (i^2(t))$	$p(t) = i(t)v(t) = Cv(t) \frac{dv}{dt}$ $= \frac{C}{2} \frac{d}{dt} (v^2(t))$
Energy	$p(t) = \frac{dw}{dt}$ $w(t) = \frac{L}{2} i^2(t)$	$p(t) = \frac{dw}{dt}$ $w(t) = \frac{C}{2} v^2(t)$

Example 1 for an RC Circuit

1. Steady-state before the switch:

2. Instantaneously after the switch:

3. In steady state after the switch:



What happens to the energy that was stored in the capacitor?

Example 1 for an RC Circuit

1. Steady-state before the switch:

$$v = 2 \text{ V}$$

$$i = 0 \text{ A}$$

$$\text{Energy stored in the capacitor} = v^2 C = 20 \mu\text{J}$$

2. Instantaneously after the switch:

$$v = 2 \text{ V}$$

$$i = -\frac{2\text{V}}{10\Omega} = -0.2 \text{ A}$$

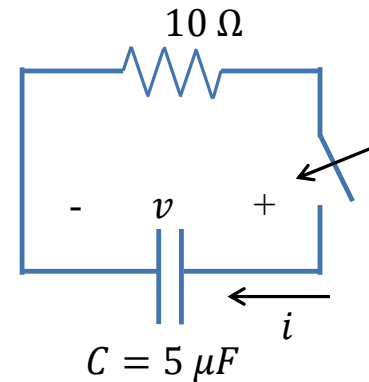
$$\text{Equation: } C \frac{dv(t)}{dt} = i(t) = -\frac{v(t)}{R}$$

3. In steady state after the switch:

$$i = 0 \text{ A}$$

$$v = 0 \text{ V}$$

$$\text{Energy stored in the capacitor} = 0 \text{ J}$$



Switch closes at
 $t = 0$ seconds

Initial voltage on
capacitor is 2 V

What happens to the
energy that was
stored in the
capacitor?

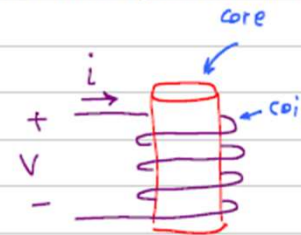
Inductor



The Inductor :

An inductor is an energy storage element composed of a coil of wire wound around a supporting core. The behavior of inductor is based on magnetic field which is created by charge in motion (current). A time varying current creates a time varying magnetic field which induces voltage in any conductor linked by the magnetic field. The parameter that relates the induced voltage to the current is called inductance, denoted by L and measured in Henrys (H).

$$L = \frac{N^2 \mu A}{l}$$

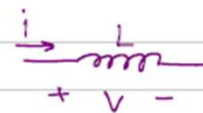


A : Cross sectional area of the core

N : number of turns

l : length of the core

μ : permeability of the core

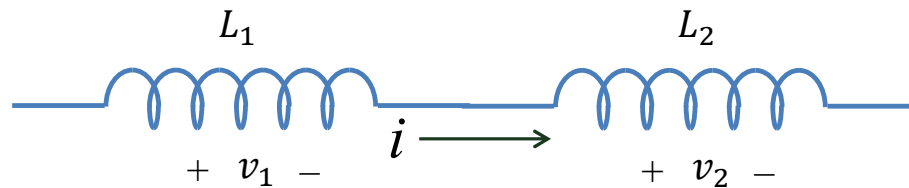


$$V = L \frac{di}{dt}$$

(if i is in the direction of voltage drop)

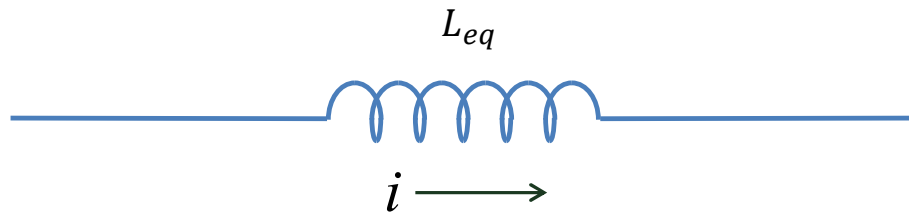
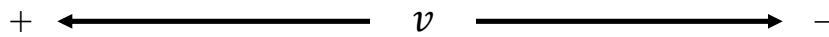
Inductors in Series

- Inductors combine in series exactly like resistors.



$$v_1 = L_1 \frac{di}{dt} \quad v_2 = L_2 \frac{di}{dt}$$

$$v = v_1 + v_2 = (L_1 + L_2) \frac{di}{dt}$$

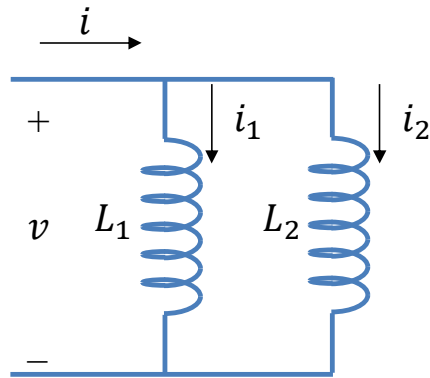


$$v = L_{eq} \frac{di}{dt}$$

$$L_{eq} = L_1 + L_2$$

Inductors in Parallel

- Inductors combine in parallel exactly like resistors.

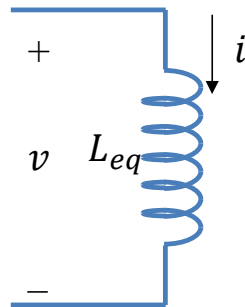


$$v = L_1 \frac{di_1}{dt}$$

$$v = L_2 \frac{di_2}{dt}$$

Using KCL: $i = i_1 + i_2$

Differentiating: $\frac{di}{dt} = \frac{di_1}{dt} + \frac{di_2}{dt}$



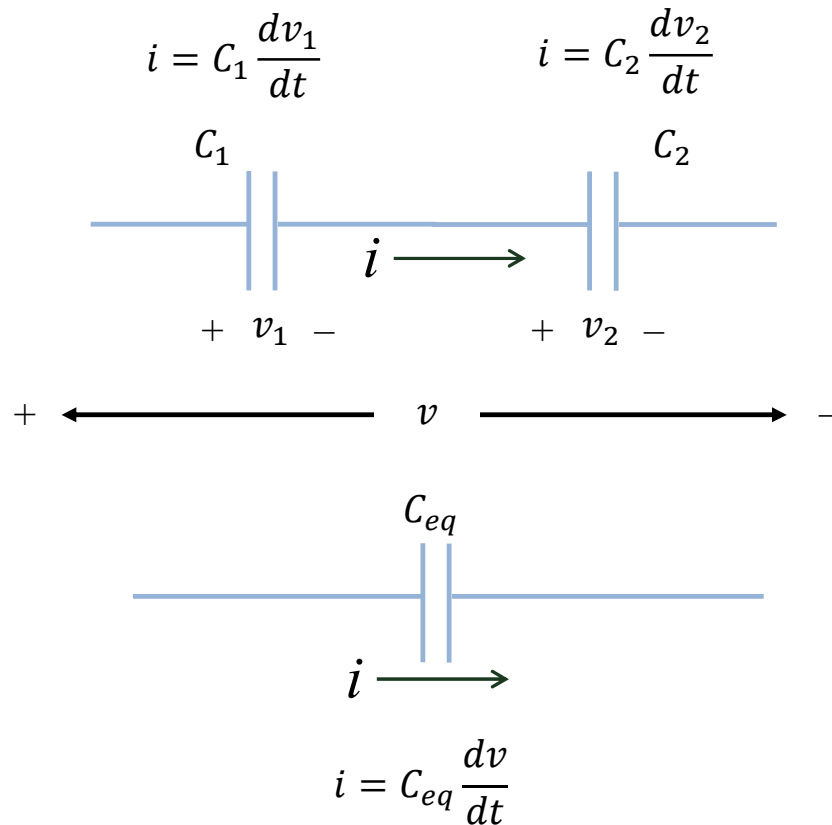
$$v = L_{eq} \frac{di}{dt}$$

Substituting: $\frac{v}{L_{eq}} = \frac{v}{L_1} + \frac{v}{L_2}$

$$\boxed{\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2}}$$

Capacitors in Series

- Capacitors in series combine like resistors in parallel.



Using KVL: $v = v_1 + v_2$

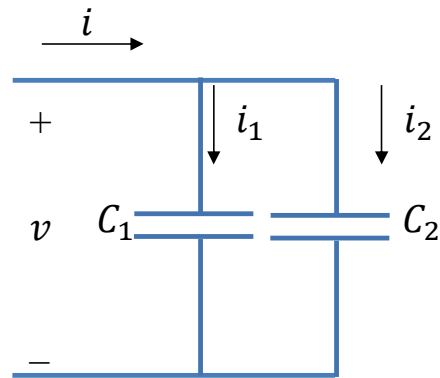
Differentiating: $\frac{dv}{dt} = \frac{dv_1}{dt} + \frac{dv_2}{dt}$

Substituting: $\frac{i}{C_{eq}} = \frac{i}{C_1} + \frac{i}{C_2}$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}$$

Capacitors in Parallel

- Capacitors in parallel combine like resistors in series.



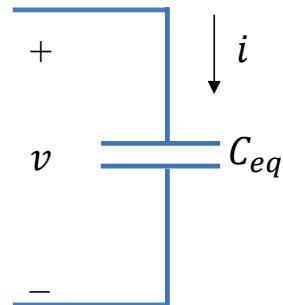
$$i_1 = C_1 \frac{dv}{dt}$$

$$i_2 = C_2 \frac{dv}{dt}$$

Using KCL: $i = i_1 + i_2$

Substituting:

$$\begin{aligned} C_{eq} \frac{dv}{dt} &= C_1 \frac{dv}{dt} + C_2 \frac{dv}{dt} \\ &= (C_1 + C_2) \frac{dv}{dt} \end{aligned}$$



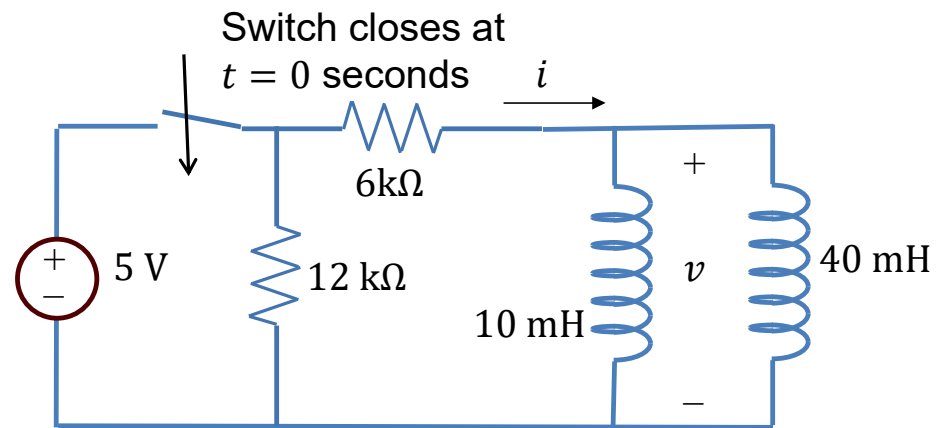
$$i = C_{eq} \frac{dv}{dt}$$

$C_{eq} = C_1 + C_2$

Example 2



- First, combine inductors in parallel



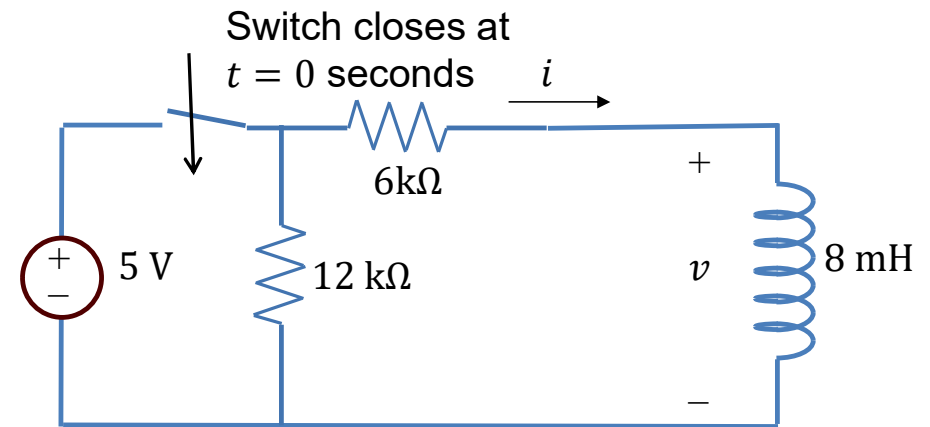
Example 2, Cont.



1. Steady-state before the switch:

2. Instantaneously after the switch:

3. In steady state after the switch:

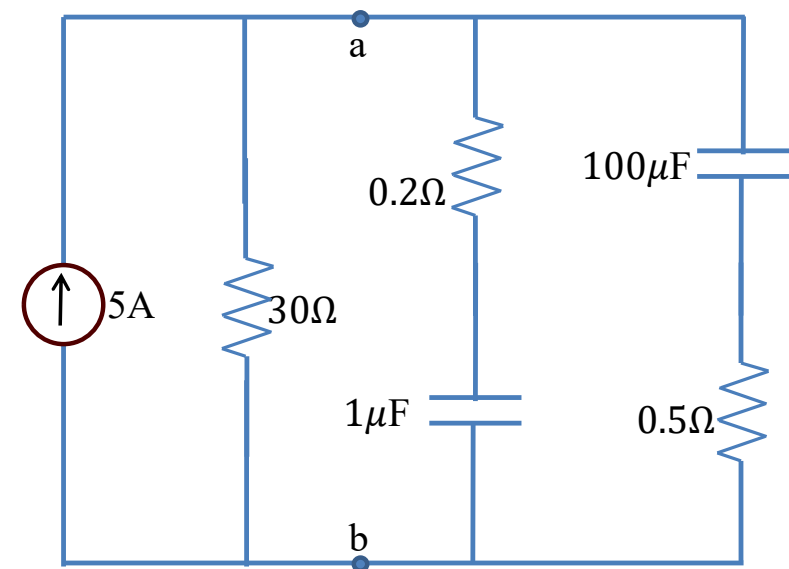


What if the switch opens back up later?

Example 3



The “switching” action is to add an accidental short circuit between a and b. What’s the instant and steady-state current through the short circuit?





What You Need to Know Today

- Capacitors
 - Fundamental equation: $i = C \frac{dv}{dt}$, where C is capacitance in Farads
 - Add in parallel, combine in series the way resistors do in parallel
 - In steady-state, they act like an open circuit
 - Energy stored is Cv^2
 - Inductors
 - Fundamental equation: $v = L \frac{di}{dt}$, where L is inductance in Henries
 - Combine in series and parallel the same way as resistors
 - In steady-state, they act like a short circuit
 - Energy stored is Li^2
 - Transient circuit solutions
 - We are solving for v and i as a function of time, $v(t)$ and $i(t)$
 - Write the differential equations using KVL and KCL
 - Next class we will start to work on solving them
-

Next Lecture

- Wednesday lecture will be posted via Zoom
- RL Circuit

